



Arielle Melamed
Consultancy

Predicting

FUTURE VOLATILITY

Introduction

Candidate Models

Evaluation Framework

Results

INTRODUCTION

GOAL: predict next 30s of volatility → supports rapid intra-day trading

$$RV = \sqrt{\sum r_i^2}$$

APPROACH: divide each 10-min interval into 30sec windows.

Dynamic behaviour → time series models

RIDGE REGRESSION

Regression

- Predicts next-period volatility from order book features

$$\hat{RV}_{t+1} = \beta_0 + \beta_1 \cdot \text{price}_t + \beta_2 \cdot \text{order}_t + \beta_3 \cdot \text{BAS}_t,$$

Ridge regularisation:

- Penalty manages multi-collinearity & prevents overfitting
- Hyperparameter tuning ($\lambda = 0.01$): λ measured against QLIKE

$$\text{minimise } \sum (RV - \hat{RV})^2 + \lambda \sum \beta_j^2$$

EWMA

Exponentially weighted moving average

- Fixed simple decay
- Recent returns matter more → feeds back → high volatility persists
- Updates quickly

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_t^2,$$

Decay Factor $\lambda = 0.94$: industry standard (RiskMetrics, 1996)

HAR-RV

Predicts future volatility directly based on returns from different time horizons

- Captures persistence well
- Such short time buckets → Not exactly intended use

$$\hat{RV}_{t+1} = \beta_0 + \beta_1 RV_t + \frac{\beta_2}{5} \sum_{i=1}^5 RV_{t-i}$$

EGARCH

ARMA-GARCH: uses returns to indirectly model volatility.

EGARCH: introduces leverage effect

- Asymmetry that bad stock performance increases volatility more than good performance

$$\log(\sigma_t^2) = \omega + \alpha \cdot g(\varepsilon_{t-1}) + \beta \log(\sigma_{t-1}^2),$$

PRE-PROCESSING & ASSESSMENT SCHEME

Pre-processing

Forward-fill missing order book rows within each time_id

- Simulate price jumps rather than artificial smoothness from interpolation.

Rolling Window Forecasting

- TRAINING fixed window: 8 minutes → PREDICT next 30s
- Window shifts forward by one bucket. Repeat cycle.
- Reflects dynamic market conditions

EVALUATION METRICS

QLIKE

- Penalises under-prediction more than over-prediction
- **Robust:** less sensitive to extreme volatility spikes (Patton 2011).

$$\text{QLIKE} = \frac{1}{T} \sum_{t=1}^T \left(\frac{RV_t}{F_t} - \log \frac{RV_t}{F_t} - 1 \right)$$

RMSE

- Penalises large errors regardless of direction → symmetric benchmark
- More sensitive to outliers

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (RV_t - F_t)^2}$$

MODEL COMPARISON

Model	QLIKE	RMSE
Ridge	0.123653	0.000499
EGARCH	0.145697	0.000546
EWMA	0.174786	0.000567
HAR	0.198399	0.000544

FINAL MODEL

**RIDGE
REGRESSION**

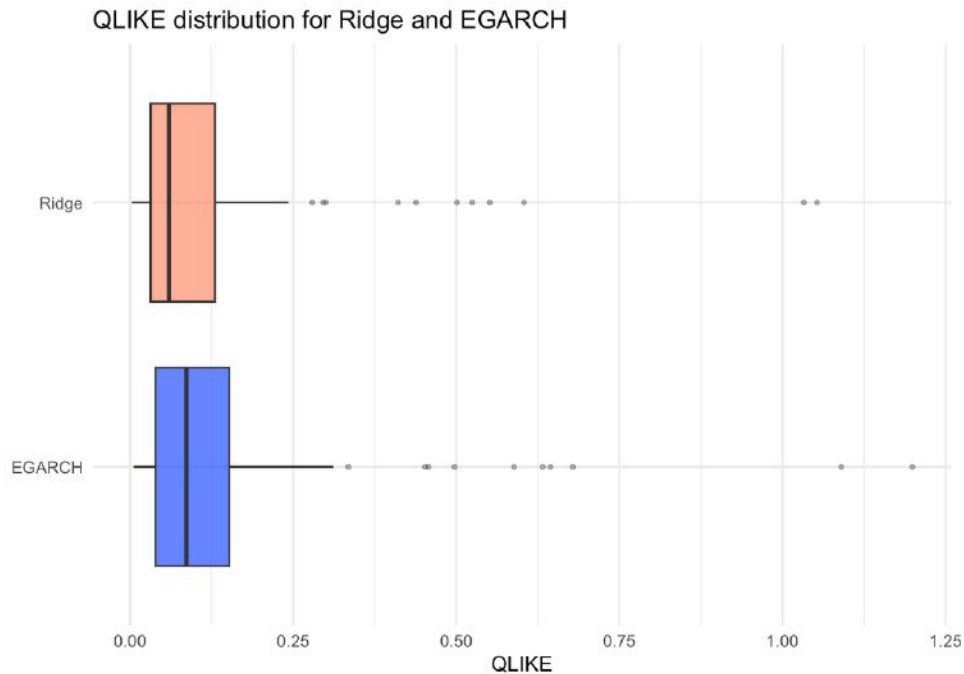
Introduction

Candidate Models

Evaluation Framework

Results

MODEL COMPARISON



FINAL MODEL

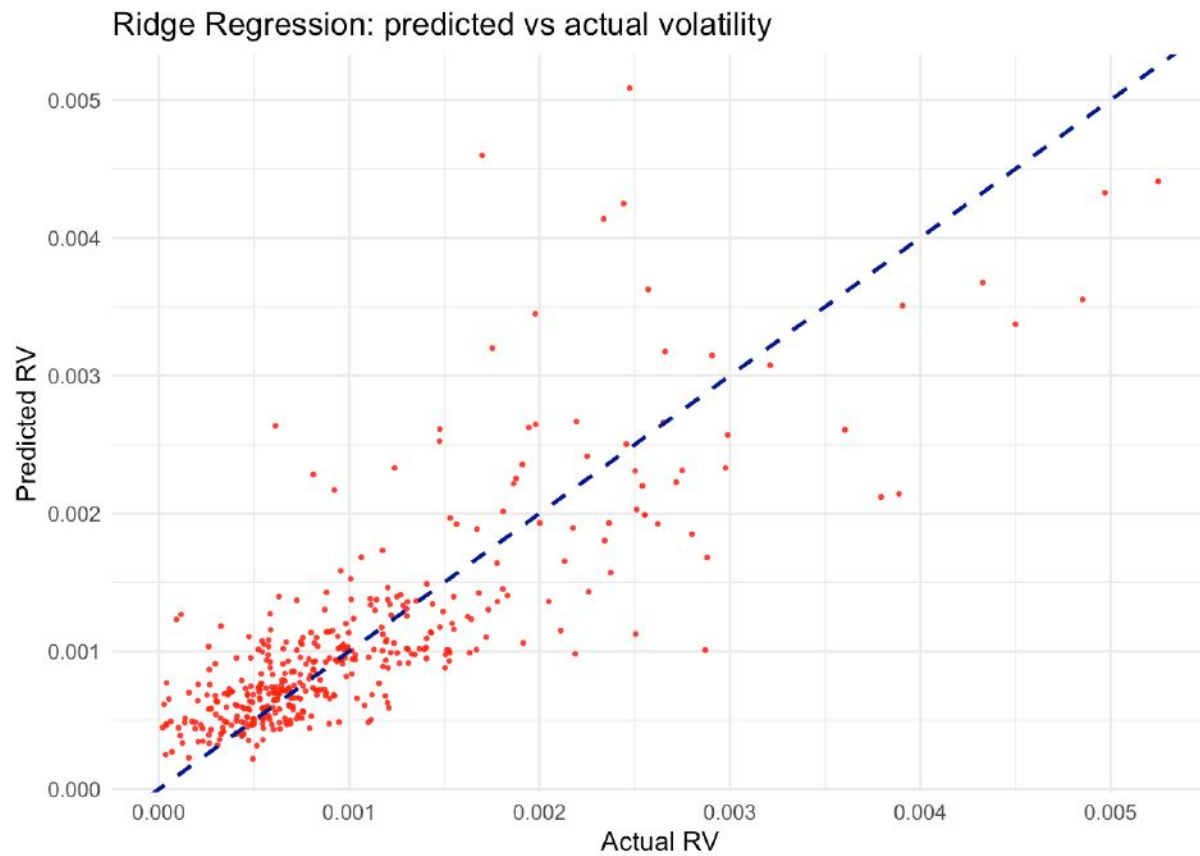
**RIDGE
REGRESSION**

Introduction

Candidate Models

Evaluation Framework

Results





Arielle Melamed
Consultancy

Thank You



REFERENCES

Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256. <https://doi.org/10.1016/j.jeconom.2010.03.034>

J.P. Morgan. (1996). RiskMetrics™ technical document (4th ed.). J.P. Morgan.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196. <https://doi.org/10.1093/jjfinec/nbp001>

Engle, R. F., & Ng, V. K. (1993). Measuring and testing the impact of news on volatility. *Journal of Finance*, 48(5), 1749–1778. <https://doi.org/10.1111/j.1540-6261.1993.tb05127.x>

I acknowledge the use of ChatGPT (<https://chat.openai.com>), version GPT-5.3, and Claude, to identify potential issues in my code and suggestions for debugging approaches.

APPENDIX: ALL MODELS

Model	QLIKE	RMSE
Ridge	0.123653	0.000499
Lasso	0.124395	0.000515
EGARCH	0.145697	0.000546
OLS	0.165113	0.000550
EWMA	0.174786	0.000567
HAR	0.198399	0.000544
WLS	0.199147	0.000541
GARCH	0.247632	0.000524

APPENDIX: LAMBDA SELECTION

